MZ2POL-02: Understanding the New Access Control Model

> **Series:** MZ2POL — Management Zone Migration: Segments, Policies, Workflows | **Notebook:** 3 of 17 | **Created:** December 2025 | **Last Updated:** 10/13/2026

Overview

This notebook provides a deep dive into the **ABAC (Attribute-Based Access Control)** framework that replaces Management Zones for access control. You'll learn how Policies, Boundaries, and Segments work together to provide flexible, scalable access management.

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Prerequisites

- Completed MZ2POL-01: Introduction
- Access to Dynatrace Account Management
- Understanding of current Management Zone configuration

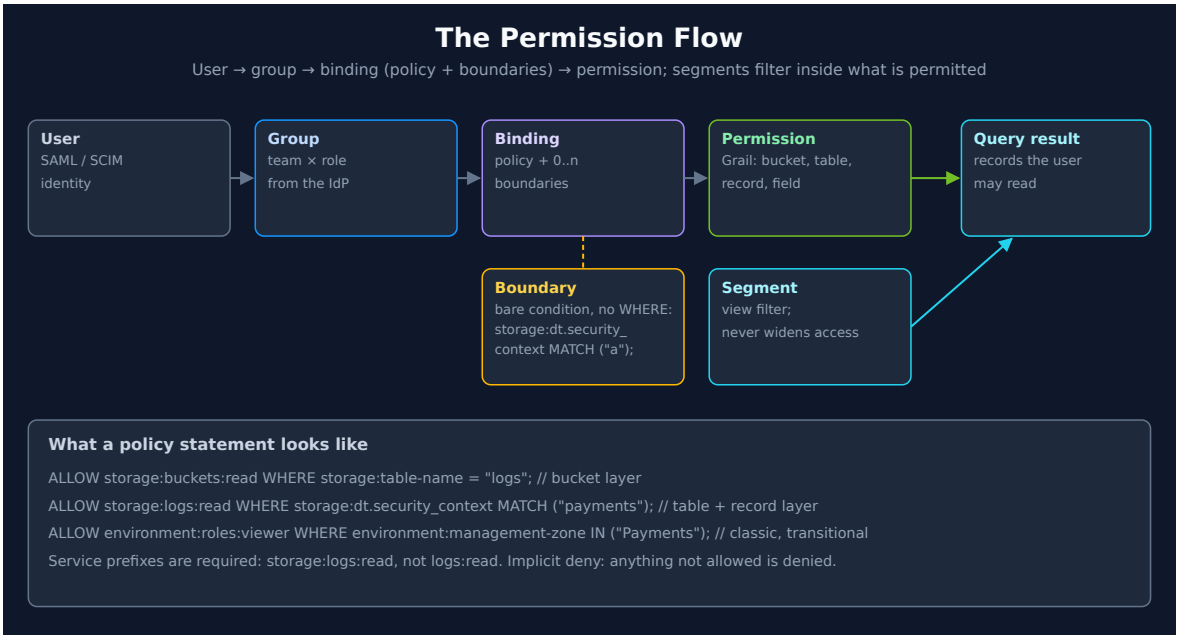
Learning Objectives

By the end of this notebook, you will:
1. Understand the ABAC framework architecture
2. Know the relationship between Policies, Boundaries, and Segments
3. Understand how permissions flow through the system
4. Be able to map MZ concepts to the new model

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1. ABAC Framework Architecture

The Permission Flow



Key Components

Component	Purpose	Configured In
Users	Individual identities	Account Management
Groups	Collections of users	Identity & Access Management
Policies	Permission definitions	Policy Management
Boundaries	Scope restrictions	Policy Boundaries
Segments	Data filtering	Segments app

How It Works Together

1. **Users** are assigned to **Groups**
2. **Groups** are bound to **Policies**
3. **Boundaries** can optionally restrict the **Policy** scope
4. **Segments** filter what data users see (independent of permissions)

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2. Policies Deep Dive

What Are Policies?

Policies are the core of ABAC - they define **WHAT** users can do.

Policy Types

Type	Description	Editable
Default Policies	Pre-defined by Dynatrace	No (read-only)
Custom Policies	Created by administrators	Yes

Default Policies Categories

> The current built-in IAM policy names are below. The classic Gen2 roles "Dynatrace Viewer / Professional User" and "Data Viewer / Editor" are no longer the IAM policy names.

Platform access policies (feature-level):
- `Standard User` - Access the environment and run Dynatrace Apps
- `Pro User` - Build, deploy, and run apps + automated workflows
- `Admin User` - Administrative access across all Platform Services

Data access policies (monitored data):
- `All Grail data read access` - Unrestricted Grail read
- `Read Logs` / `Read Spans` / `Read Metrics` / `Read Events` / `Read Entities` / `Read BizEvents` / `Read System Events` - Granular per-table read

> **`Settings Reader` / `Settings Writer` are legacy.** The default-policies reference lists them, with *Storage Default Monitoring Read* and *Storage All System Data Read*, under **Legacy default policies**: *"They are not accessible for new policy assignments, but existing assignments of these policies remain until removed."* For settings access, write a custom policy with `settings:objects:read` or `settings:objects:write` (plus `settings:schemas:read`), scoped by `settings:schemaId`, `settings:schemaGroup` or `settings:dt.security_context`.
>
> **Sources:** Dynatrace default policies reference (DT docs), IAM policy reference (DT docs) — `settings:objects:write`: *"Enables writing of settings objects belonging to the schema"*.

Policy Statement Structure

```
ALLOW &lt;service&gt;:&lt;resource&gt;:&lt;action&gt; [WHERE &lt;condition&gt;]
```

Examples:

```
ALLOW storage:buckets:read
ALLOW settings:objects:read WHERE settings:schemaId =
"builtin:alerting.profile"
ALLOW storage:logs:read WHERE storage:dt.security_context = "team-a"
```

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3. Boundaries Deep Dive

What Are Boundaries?

Boundaries restrict **WHERE** policies apply - they limit the scope of permissions.

Key Characteristics

- **Optional** but powerful for fine-grained access control
- Work **together** with policies (not standalone)
- **Further restrict** existing policy permissions
- Enable **reusability** across multiple policy assignments

Boundary Query Syntax

A boundary query is a **bare condition expression** — no `WHERE` keyword (that belongs to policy statements), one condition per line:

```
&lt;field&gt; &lt;operator&gt; "&lt;value&gt;"
```

Supported Operators — they vary by attribute (*"Not every operator applies to every service attribute."*):
- Grail record fields (`storage:dt.security_context`, `storage:dt.host_group.id`, `storage:k8s.namespace.name` …): `=`, `IN`, `startsWith`, `MATCH` — **no negation**
- `storage:bucket-name`: all of the above plus `!=`, `NOT IN`, `NOT startsWith`
- `environment:management-zone`: `=`, `!=`, `IN`, `startsWith`, `NOT startsWith`, `MATCH`

> **Sources:** IAM policy statement syntax and examples (DT docs), IAM policy reference (DT docs) — per-attribute operator lists.

Common Fields:

- `environment` - Environment restrictions
- `environment:management-zone` - MZ-based restrictions (Classic)
- `storage:dt.security_context` - Security context filtering
- `storage:bucket-name` - Bucket-based restrictions

> **Note:** `MATCH` provides wildcard pattern matching on Grail record fields (and on `environment:management-zone`) — `storage:dt.security_context MATCH("team-*")` — and supports `*` at any position (unlike `startsWith`). It is also the operator to use whenever a record field can hold an **array**: *"Using =, STARTSWITH or IN when the field holds an array will always return false."* Events carry `dt.security_context` as an array (MZ2POL-00 §5), so write `MATCH` for it in every policy and boundary (Permissions in Grail (DT docs)). For boundary design using structured multi-dimensional context values (`comp/bu/app` for transversal team access), see **MZ2POL-04: Policies and Boundaries** and **IAM-05: Boundary Design**.

Boundary Examples

Restrict to specific Management Zone (transitional):

```
environment:management-zone = "Production-NA"
```

Restrict by Management Zone prefix:

```
environment:management-zone startsWith "mgmt_na"
```

Restrict by Security Context:

```
storage:dt.security_context = "team-frontend"
```

Restrict by Bucket (compliance case only — §4):

```
storage:bucket-name IN ("pci_logs", "pci_spans")
```

Boundary Limitations

Limitation	Workaround
Max 10 conditions per boundary	Create multiple boundaries
No logical operators — one line per condition	Repeated lines on the **same** condition multiply the statement (OR); lines on different conditions each scope their own permissions. Several boundaries on one binding are evaluated separately (MZ2POL-12 §4)
Bare conditions (no `WHERE`)	`WHERE` belongs to policy statements, not boundaries

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4. Buckets: Data Partitioning, Not the Access Mechanism

What Are Buckets?

Buckets are Grail storage containers that partition your data. A management zone that restricted access does **not** translate into a bucket by default: Dynatrace states that buckets are *"not intended as the primary mechanism for securing or segmenting data access."* Access is a record-level boundary on `dt.security_context` (**MZ2POL-12: IAM Reference Architecture**; **ORGNZ-06: Security Context**).

What Buckets Are For

Use	Description
Retention control	Different retention periods need different buckets
Cost allocation	Storage and query consumption attributable per bucket
Query optimization	A query scoped to a smaller bucket scans less data
Specific compliance cases	A hard exclusion — for example a `DENY` on one bucket — that record-level conditions cannot express

The partition-data page says buckets are *"best used for organizing data by retention time, cost allocation, or query optimization"*, and recommends starting from IAM policies, with buckets *"to play a supporting role for very high-level separation."* The one access job only a bucket can do is a `DENY`: *"Such access control is only possible with buckets."* **ORGNZ-05: Bucket-Level Access Control** covers when that scenario fits.

> **Sources:** Partition data (DT docs) — *"not intended as the primary mechanism for securing or segmenting data access."*

Buckets vs Boundaries vs Segments

Aspect	Buckets	Boundaries	Segments
Isolation	Physical separation	Query-time filtering	Query-time filtering
Reversible	❌ No (data can't move)	✅ Yes	✅ Yes
Performance	Best (smaller data sets)	Good	Good
Flexibility	Low (plan carefully)	High	High
Use case	Retention, cost, query scope; specific compliance exclusions	Permission scoping (team access)	Data filtering

Bucket Limitations

> ⚠️ **Critical:** Plan buckets carefully - these constraints cannot be changed later.

Limitation	Details
One data type per bucket	Logs, metrics, events, OR spans - not mixed
Names are immutable	Bucket names CANNOT be changed after creation
No data migration	Data CANNOT be moved between buckets

Limitation	Details
Naming rules	3-100 chars, lowercase alphanumeric, underscores, hyphens only
Maximum buckets	80 per environment (default limit)
Optimal ingest	~1 TB/day per bucket for best query performance
Acceptable ingest	1-3 TB/day per bucket (limited query window)
Maximum ingest	No hard per-bucket cap is documented — Dynatrace describes ingesting *"well over 15 TB per day into a single bucket"* when queries stay within narrow timeframes. A larger bucket costs queryable window per scan, and Dynatrace pages give different split points (Partition data (DT docs), Logs bucket strategy (DT docs))

Default Query Setting That Affects Bucket Sizing

Setting	Default	Impact
`scanLimitGBytes` on `fetch`	500 GB of uncompressed data (`-1` reads everything in the timeframe)	A larger bucket covers a shorter time window per default scan

> **Sources:** Data source commands (DT docs) — *"The default value is 500GB unless specified otherwise."*

Using Buckets in Policies

Only where one of the scenarios above applies — for example, a compliance bucket that only one group may read. A bucket permission is needed on top of the table permission: `storage:buckets:read` is *"Required additionally to a table permission."*

```
// Policy: a compliance case – access to dedicated buckets
ALLOW storage:buckets:read WHERE storage:bucket-name IN ("pci_logs",
"pci_spans");
ALLOW storage:logs:read WHERE storage:bucket-name = "pci_logs";
ALLOW storage:spans:read WHERE storage:bucket-name = "pci_spans";
```

Team Access Uses a Boundary, Not a Bucket

The team boundary scopes records by security context, not by bucket. A boundary is **bare condition lines, each ending in `;`** — no `ALLOW`, no `WHERE`. **One boundary can cover every domain the bound policies grant** — Grail lines and, while classic pages are used, the zone line (MZ2POL-04 §6, MZ2POL-12 §6):

```
// Boundary "team-frontend", bound with Standard User + Read Logs / Read Spans
/ Read Metrics / Read Events
storage:dt.security_context MATCH ("frontend", "shared");
environment:management-zone IN ("Frontend-Team");
```

The second line is transitional: it scopes the classic `environment:roles:viewer` permission inside Standard User while classic pages are still in use. Add a `storage:bucket-name` line only in a compliance case from the table above.

> **`MATCH` is not Gen3-only** — the IAM policy reference lists it for `environment:management-zone` too. `storage:entities:read` is a Grail permission that accepts only `storage:dt.security_context` and `storage:entity.type`; `MATCH("*")` on any attribute matches everything.
>
> **Sources:** IAM policy reference (DT docs) — `environment:roles:viewer`: *"Grants user the Access environment permission."*; `storage:buckets:read`: *"Required additionally to a table permission."*

When to Use Buckets vs Other Options

Scenario	Recommended Approach
Team data access (the usual access-control zone)	**Boundary** on `dt.security_context` (MZ2POL-12)
Different retention periods or cost attribution	**Buckets**
Compliance case that needs a hard exclusion (`DENY` on a bucket)	**Bucket** + bucket condition, on top of the security-context boundary (ORGNZ-05)
Ad-hoc data filtering	**Segments**

MZ to Bucket Mapping

MZ Pattern	Bucket Strategy
Team MZs	No bucket by default — a security-context value per team
Environment MZs	A bucket only if the environments need different retention
Regional MZs	Usually none — a segment, or a security-context value if access differs
Compliance MZs	A compliance bucket where retention or a hard exclusion requires it

> **For comprehensive bucket guidance**, see **ORGNZ-03: Bucket Strategy and Design**; for record-level access, **ORGNZ-06: Security Context** and **MZ2POL-12**.

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5. Segments Deep Dive

What Are Segments?

Segments are **DQL-based filter conditions** that control what data users see - they're the replacement for MZ data filtering.

Key Characteristics

- **Query-time evaluation** — conditions are injected into the query
- **Layerable** - several segments can be applied together
- **DQL-powered** - full query language flexibility
- Support **variables** for dynamic filtering
- **Independent of permissions** - filtering only

How Segments Work in DQL

When a segment is applied, Grail:
1. Evaluates segment conditions relevant to the query
2. Applies filters based on the targeted data object
3. Multiple conditions for same data object = OR combined

Segment vs. Management Zone Filtering

Aspect	Management Zone	Segment
Evaluation	Rules assign entities to the zone; dimension rules match logs and metrics	Conditions injected into each query
Grail data	Not evaluated	Native
Flexibility	Rule conditions	DQL filter conditions
Variables	No	Yes
Combining dimensions	Within one zone's rules	Layer several segments

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6. Querying Current Access Configuration

View Services with Security Context

Security Context is key for access control in the new model:

```
// List services and their security context
// Security context is used for fine-grained access control
fetch dt.entity.service
| fields entity.name,
         dt.security_context,
         managementZones
| filter isNotNull(dt.security_context)
| sort entity.name asc
| limit 50

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
```

Analyze Entity Types and Their Attributes

Understanding entity attributes helps design effective segments:

```
// Analyze host entity attributes for segment planning
// Tags and metadata are useful for segment conditions
fetch dt.entity.host
| fields entity.name,
         tags,
         managementZones
| limit 20

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

Check Kubernetes Cluster Distribution

K8s clusters often map to organizational boundaries:

```
// List Kubernetes clusters with their attributes
// Clusters often align with team or environment boundaries
fetch dt.entity.kubernetes_cluster
| fields entity.name,
         tags,
         managementZones
| sort entity.name asc

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

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7. Mapping MZ Concepts to New Model

Common MZ Patterns and Their Replacements

MZ Pattern	New Approach
Team-based MZs	Security Context + Policies
Environment MZs (Dev/Prod)	Boundaries on a record field that carries the environment — a `dt.security_context` value such as `prod/payments`, or a built-in field
Region MZs	Segments with cloud region filters
Application MZs	Segments with service/app filters
Multi-tenant MZs	Boundaries + Security Context

Example: Team-Based Access Control

Old (Management Zone):
- MZ: "Team-Frontend" with rules for frontend services
- Users assigned to MZ get filtered view

New (Policies + Boundaries + Segments) — the MZ2POL-12 §6 shape:
1. **Policies**: `Standard User` + `Read Logs` / `Read Spans` / `Read Metrics` / `Read Events`. Standard User alone grants no table read — its Grail lines cover bucket and fieldset definitions and segments, not `storage:logs:read`.
2. **Boundary** `team-frontend`, the only boundary on the binding:
`storage:dt.security_context MATCH ("frontend", "shared");`
`environment:management-zone IN ("Team-Frontend");`
The zone line is transitional. Standard User also grants the classic `environment:roles:viewer`, and without that line the classic permission would be unscoped.
3. **Segment**: `primary_tags.team = "frontend"` as a convenience filter — it changes the view, not the permission.

> **Sources:** Dynatrace default policies reference (DT docs) — Standard User: *"ALLOW environment:roles:viewer, environment:roles:view-security-problems;"*

Example: Environment Separation

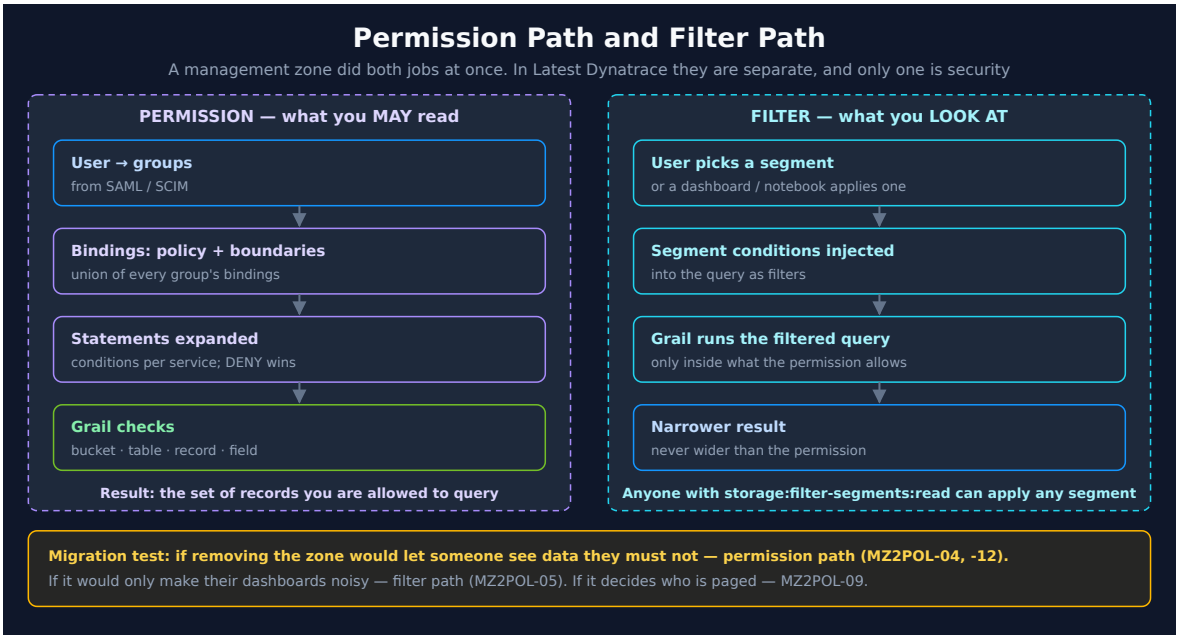
Old (Management Zone):
- MZ: "Production" with host/service rules
- MZ: "Development" with different rules

New (Policies + Boundaries + Segments):
1. **Policy**: Same policy for both groups
2. **Boundary (Prod)**: Environment-specific restrictions
3. **Boundary (Dev)**: Environment-specific restrictions
4. **Segments**: Environment-based data filters

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8. Access Control Decision Flow

Permission and Filter Are Separate Paths



The permission path follows a handful of documented rules — implicit deny, a union across every binding, boundaries applied separately and only to their own service, `DENY` overriding. They matter more in a migration than any syntax detail, because each one can make a translated zone wider than the original. **MZ2POL-12 §3** walks through them with Dynatrace's wording.

> **Sources:** Policy boundaries (DT docs) — *"Only conditions added to service configuration will be applied to relative permissions."*, Visibility of segments (DT docs).

Key Differences from MZ

MZ Behavior	New Behavior
Single construct for access + filtering	Separate concerns (Policies vs Segments)
Zone membership assigned to entities by rules	Conditions evaluated on each record at query time
Entity rules, plus dimension rules for logs and metrics	Any field on the record

MZ Behavior	New Behavior
Flat structure	Hierarchical (groups → policies → boundaries)

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9. Best Practices for the New Model

Policy Design

1. **Start with default policies** - customize only when needed
2. **Use least privilege** - grant minimum required permissions
3. **Group similar permissions** - avoid policy sprawl
4. **Document policy purpose** - maintain clarity

Boundary Design

1. **Create reusable boundaries** - one boundary, many uses
2. **Use meaningful names** - indicate scope clearly
3. **Keep conditions simple** - easier to audit
4. **Leverage security context** - for entity-level control

Segment Design

1. **Align with business structure** - teams, regions, products
2. **Use variables** - for dynamic, flexible filters
3. **Test thoroughly** - verify filtering works as expected
4. **Layer segments** - combine for precise filtering

Summary

In this notebook, you learned:

1. **ABAC Framework**: How Users → Groups → Policies → Permissions flow

2. **Policies**: Define WHAT users can do (permissions)
3. **Boundaries**: Restrict WHERE policies apply (scope)
4. **Segments**: Filter WHAT data users see (DQL-based)
5. **Mapping**: How MZ patterns translate to the new model

Next Steps

Continue to **MZ2POL-03: Assessment and Migration Planning** to:
- Audit your current MZ configuration in detail
- Create a migration mapping document
- Plan the phased migration approach

Additional Resources

- Working with Policies
- IAM Policy Reference
- Default Policies Reference
- Grant Access to Entities with Security Context

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